

CURRICULUM VITAE

Dr Muhammad Usman

Postdoc in Cyber Security & Machine Learning (University of Surrey, UK) Ranked 285 in QS Ranking 2025
PhD – ICT – Cyber Security (Griffith University, Australia), Ranked 255 in QS Ranking 2025
MS (Computer Science), BS (Computer Science), Pakistan
Fellow Advance HE, UK, Member IET, UK, Member BCS, Member IEEE,
Secretary-Elect Global and Member IEEE CTSoc Technical Committees Member
Network Security Research Group, Griffith University, Australia
Member Computer Science Teachers Association endorsed by ACM, USA Certified Networking and Security Specialist, Juniper Networks, USA
Research Group Leader: HCI-S²: Human-Centred Intelligent and Secure Systems

Weblink: <https://www.linkedin.com/in/dr-muhammad-usman-947b9b30/>

Address:

Dr Muhammad Usman
23 Broxton Avenue,
BL3 3TG, Bolton, United Kingdom
email: manilasani@yahoo.com

Personal Information

Date of Birth	24 January 1983 Male
Gender	United Kingdom
Nationality	

Brief Bio:

Dr Usman received **MS (Computer Science)** degree with **specialisation in Cyber Security**, with **first class**. He completed his **PhD from the School of ICT (Cyber Security)**, Griffith University, Australia. He was **Postdoc (Research Fellow) in Cyber Security and Machine Learning**, University of Surrey, UK.

Dr Usman has nearly **22 years of all-round academic and industry experience**. He has worked in different continents of the world: **Australia, Asia and Europe** and have developed an extensive network of research and academic collaborators from across the globe. His experience is evident from his **diverse set of industry** (Network and Hardware Engineer and System Administrator), **academic** (Instructor, Lecturer, Assistant Professor (Computer Science), Senior Lecturer in Cyber Security and Senior Lecturer in Cyber Security & Forensics), **research** (Research Fellow/ Postdoc) **and managerial positions** (project manager).

He **provides advisory and consultancy** in multiple areas such as higher education; curriculum design & development; cyber security, digital forensics; research and development; and career professional development trainings. He has been a member of **COVID 19 Outbreak Expert Database of UK Parliament** and has provided **advisory services to Govt. of Pakistan**.

Dr Usman is currently **leading a research group**: [Human-Centred Intelligent and Secure Systems](#). His current research interests include interdisciplinary areas around cyber security, digital forensics; trust & privacy; AI-driven complex systems; human-centred systems; digital twins; Agritech; Healthtech; community tech; consumer electronics; drones, electrical and autonomous vehicles; and formal and statistical modelling. He has authored and co-authored **more than 50 published research articles and a book**. A significant proportion of his research articles has **appeared in top venues such as in numerous IEEE Transactions, IEEE IOT Journal, ACM Computing Surveys, Elsevier Computer Networks, several Q1 journals, and a prestigious conference: Local Computer Networks**. One of his research papers has received the **best paper award** in the conference IEEE ComTech 2017.

Dr. Usman has been **recipient of several research fundings**. Notably he has **led both Transnational Curricula for Digital Forensics (2023 -2024)** and **Going Global Partnership PAK-UK Education Gateway Mobility Partnership for Faculty (2023 -2024)** as **Principal Applicant (PI)**, funded by **British Council**. He has also **led Design and Analysis of Web Services Ranking Methods for Internet of Things (2017-2018)**, as **PI**, funded by **Higher Education Commission of Pakistan**. He has been **Co-I** of the project: **Framework for AI-enabled Drone Forensics Analysis (2023-2024)**, funded by **NAUSS Saudi Arabia**. He also **co-led three projects**: (i) **Code Vulnerability Scanning for Java Language**; (ii) **Code Vulnerability Scanning for C Language** and (iii) **Data Leak Prevention (2017-2018)**, funded by an overseas public sector organisation.

Dr Usman has **growing strong international research reputation** which is evident from several indicators such as continued to be invited for **keynote talks in conferences and talk series**; **guest editorial roles**; **editorial membership of a journal**; **general/ co-general chair of international colloquiums and symposiums**; **steering committee memberships**; **member technical programme committees**; **session chair**; **PhD viva chair**; **grant reviewer (UKRI, EPSRC, British Council)**; **examiner for PhD theses**; **leading a faculty delegate from different universities of the UK to Pakistan on faculty exchange programme**; **General Chair of Hackathon**; **elected as a member of Security and Privacy of CE Hardware and Software Systems (SPC) Technical Committee (TC) by IEEE CTSoc Consumer Technology Society**; and a **reviewer for esteemed journals**. He has served in different leading capacities, such as a **focal person**, **publication Chair**, **organising committee member and/or TPC member** of several IEEE conferences.

He is **Programme Leader** for [BSc \(Hons\) Computing \(Networks, Cyber Security and Forensics\)](#). He is also a **member of Computer Science Teachers Association**, endorsed by Association for Computing Machinery, USA. He is a **Juniper certified networking and security specialist**, and **member of IEEE and BCS**.

Education

Doctor of Philosophy (Ph.D.)		Completion year: 2015
Major Research Centre Research Group School	Computer Networks and Security Institute for Integrated and Intelligent Systems (IIIS) Network Security School of Information and Communication Technology, Griffith University, Gold Coast campus, Australia Design and Analysis of Mobile Agent-Enabled Anomaly Detection and Verification in Wireless Sensor Networks	
Research Topic		
M. Science (Computer Science)		Completion year: 2009
MS / M.Phil Major Institution	Computer Science Network Cyber Security University Institute of Information Technology (UIIT) Pir Mehr Ali Shah, Arid Agriculture University, Rawalpindi, Pakistan Secure and Seamless Vertical Handoff in Integrated Wireless Networks	
Research Thesis		
B. Science (Computer Science)		Completion year: 2006
BS Major Institution	Computer Science Computer Science Allama Iqbal Open University, Islamabad, Pakistan	

Professional Certifications

Intel Teach Program	2009
Intel Corporation, USA	
JNO-350, Certified Internet Specialist, (JNCIS-ER)	2009
Juniper Networks, California, USA	
JNO-330, Certified Internet Specialist (JNCIS-JES)	2009
Juniper Networks, California, USA	
JNO-400, Certified Internet Associate, (JNCIA-EX)	2009
Juniper Networks, California, USA	
JNO-342, Certified Internet Associate (JNCIA-ER)	2009
Juniper Networks, California, USA	
Database Management Systems (DBMS)	2007
Quaid-I-Azam University, Islamabad, Pakistan	
Computer Networking	2006
Quaid-I-Azam University, Islamabad, Pakistan	
Linux System Administration	2006
Pakistan Computer Bureau, Islamabad, Pakistan	
E-Commerce (Webpage Designing)	2003
Trade Training Centre, Islamabad, Pakistan	

Government Advisory Roles:

- Member of UK Government COVID-19 Database (from 2021)
- Member of Champions of Reforms Group, Gov.t of Pakistan (December 2022 – to date)
- Pre-CDWP Meeting by M/o PD & SI, Pakistan (26 January 2023)

Honors and Awards

Best Research Paper Award in IEEE ComTech	2017
Research paper titled as "Dynamic Trust Evaluation of Learning objects" co-authored with Ms. Kanwal Mahmud won the best paper award in the IEEE ComTech 2017 conference.	
Griffith University International Postgraduate Research Scholarship (GUIPRS) for PhD	2012-2015
Honoured with highly competitive GUIPRS (tuition fee) scholarship to pursue PhD studies from a well accredited Australian university	
Griffith University Postgraduate Research Scholarship (GUPRS) for PhD	2012-2015
Honoured with highly competitive GUPRS (living stipend) scholarship to pursue PhD studies from a well accredited Australian university	
Conference Travel Grant – WoWMoM 2014	2014
Awarded with conference travel grant amounting to 906AUD by School of ICT, Griffith University, Australia in Semester 1, 2014	
Conference Travel Grant – 38th LCN	2013
Awarded with conference travel grant amounting to 450AUD by School of ICT, Griffith University, Australia in Semester 2, 2013	
1 st Position in Computer Networks Group of MS (CS)	2006-2009
1st position in Computer Networks Group in MS (CS)	
Vice Chancellor Merit Scholarship	2 nd Semester of MS (Computer Science)
Vice Chancellor Merit Scholarship in 2nd Semester of MS (CS) by scoring 4.0/4.0 GPA	
Academic Standing	Last three semesters of MS (Computer Science)
Scored three consecutive 4/4 GPAs in last three semesters of MS (CS)	

Research & Development Grants

Submitted:

1. HeMRoC: Heterogeneous Multi-Robot Coordination Framework for Inspection in Constrained Spaces (submitted in March 2025) 2026 - 2028
 - Dr. Muhammad Usman Co-Investigator (UK) – Leading WP4
 - Funder: UKRI
 - The funding application is submitted by a consortium of 3 higher education institutes and two industry partners from the UK
 - Amount: approximately 3 million GBP
2. A Knowledgebase to Counter Seriously Organised Crime Anti-Forensic Evasion Using AI (submitted in February 2025)
 - Dr. Muhammad Usman Principal-Investigator (UK)
 - Funder: National Crime Agency
 - The funding application is submitted by 10 colleagues from Law, criminal justice and psychology departments, EHU
 - Amount: approximately 100,000 GBP

Successful:

3. TRAPCAT: TRansformative Artificial intelligence research for Personalised CAncer Treatment (successful) 2025 - 2026
 - Dr. Muhammad Usman Co-Investigator (UK)
 - Funder: British Council
 - The project application was submitted by a consortium of 5 higher education institutes and associated partners from UK and Egypt Amount: 79,908 GBP

4. A Framework for AI-enabled Drone Forensics Analysis (Successful)
 - Dr. Muhammad Usman Co-Investigator (UK)
 - NAUSS Saudi Arabia
 - 2023 - 2024
 - The project application submitted and funding received by a consortium of 3 higher education institutes from Saudi Arabia, UK and Canada
 - Amount: 60,800 Saudi Riyals
5. Going Global Partnership PAK-UK Education Gateway Mobility Partnership for Faculty (Successful)

2023 - 2024

 - **Dr. Muhammad Usman** Principal Investigator (UK)
 - British Council
 - The project application submitted and funding received by a consortium of 16 higher education institutes from UK and Pakistan
 - Amount: £65,000 approximately
6. Transnational Curricula for Digital Forensics (Successful)

2023 - 2024

 - **Dr. Muhammad Usman** Principal Investigator (UK)
 - British Council
 - The project application submitted and funding received by a consortium of 2 higher education institutes, i.e., EHU, UK and NAUSS, Saudi Arabia
 - Amount: £45,000
7. SPEAR: Secure and PrivatE smart gRid (Successful) 2018 -

2019

 - **Dr. Muhammad Usman** Research Fellow with Principal Investigator, Dr. Emmanouil (Manos) Panaousis (University of Surrey, United Kingdom)
 - A Multimillion Euros European Union Funded Project under the H2020 program
 - The project is being run by a consortium of 17 (7 higher education and 10 industry) partners from 11 European countries
 - Amount: Multimillion Euros
8. Design and Analysis of Web Services Ranking Methods for Internetof Things (Successful – did not obtained due to residence country) 2017 -

2018

 - **Dr. Muhammad Usman**, Principal Investigator, and Dr. Onaiza Maqbool, Co-Principal Investigator
 - Amount: PKR 0.477 million
 - Funded by Higher Education Commission of Pakistan
9. Data Leak Prevention (Successful) 2017 -

2018

 - Dr. Muddassar Azam Sindhu, Principal Investigator, **Dr. Muhammad Usman**, Co-Principal Investigator, and Dr. Onaiza Maqbool, Co-Principal Investigator
 - Amount: PKR 0.15 million
 - Funded by a Public Sector Organization
10. Code Vulnerability Scanner for Java Language (Successful) 2017 -

2018

 - Dr. Muddassar Azam Sindhu, Principal Investigator, **Dr. Muhammad Usman**, Co-Principal Investigator, and Dr. Onaiza Maqbool, Co-Principal Investigator
 - Amount: PKR 0.15 million
 - Funded by a Public Sector Organization
11. Code Vulnerability Scanner for C Language (Successful) 2017 -

2018

 - Dr. Muddassar Azam Sindhu, Principal Investigator, **Dr. Muhammad Usman**, Co-Principal Investigator, and Dr. Onaiza Maqbool, Co-Principal Investigator
 - Amount: PKR 0.15 million
 - Funded by a Public Sector Organisation

Unsuccessful:

12. RITICS Fellowship: Trust-Centric Cognitive Security in Software Supply Chain for Cyber-Physical Systems

(unsuccessful)

2024

- **Dr. Muhammad Usman** Principal Investigator (UK)
- Funder: RITICS - UKRI
- The research fellowship submitted with interdisciplinary advisors from 3 higher education institutes,
- Amount: **49,843 GBP**

13. SkillUP North-West: Cyber Security Skills for Future Leaders and SMEs (EOI accepted, proposal unsuccessful)

2024 - 2025

- **Dr. Muhammad Usman** Principal-Investigator (UK)
- Funder: Innovate UK
- The project application submitted by a consortium of 3 higher education institutes
- Amount: **150,000 GBP**

14. A Network+ for Bridging the Technology Translation Gap for Independent Living of People with Dementia (unsuccessful)

- **Dr. Muhammad Usman** Co-Investigator (UK)
- Funder: Innovate UKRI
- The project application submitted by a consortium of 7 higher education institutes
- Amount: **1.57 million GBP**

15. Prosumer-Centric Intelligent and Trustworthy Framework for Clean Energy Trading (unsuccessful)

- **Dr. Muhammad Usman** Principal Investigator (UK)
- Funder: British Council
- The project application submitted with 8 higher education institutes from Saudi Arabia and UK Amount: **40,000 GBP**

16. Human-Centred Next Generation AI for Sustainable Agriculture (unsuccessful)

- **Dr. Muhammad Usman** Co-coordinator (UK) – Leading WP6 and technical lead of the project
- Funder: EU HorizonThe project application submitted by a consortium of 10 higher education institutes from the UK, Ireland, Italy, Germany, Spain, Pakistan, China and Korea
- Amount: **2,981,422.50 Euros (EHU Share: 450,857.50 Euros)**

17. A Secure, Privacy-preserving and Transparent Framework for Shrimp Mortality Risk Prediction and Data Sharing in Vietnam for Enhanced Quality Assurance (unsuccessful)

2024 - 2025

- **Dr. Muhammad Usman** Co-Investigator (UK)
- Funder: British Council
- The project application submitted by a consortium of 4 higher education institutes and associated partners from UK and Vietnam
- Amount: **78,896 GBP**

18. Artificial Intelligence-Augmented End-Tidal carbon dioxide Monitoring for Precision Diagnostics and Prevention of Respiratory Ailments in Pakistan (unsuccessful)

2024 - 2027

- **Dr. Muhammad Usman** Co-Investigator (UK) – Lead WP4
- Funder: UKRI (Global Health Call)
- The project application submitted and funding received by a consortium of 3 higher education institutes from Saudi Arabia, UK and Canada
- Amount: **692,465 GBP (EHU Share: 322414 GBP)**

- **Dr. Muhammad Usman** Principal Investigator (UK)
- Funder: British Council
- The project application submitted by a consortium of 12 higher education institutes from UK and Pakistan
- **Amount: 110,000 GBP approximately**

Experience History/ Work Experience

Senior Lecturer in Cyber Security & Forensics / Programme Leader for BSc Computing (Networks, Cyber Security and Forensics) / Research Group Leader: HCIS²: Human-Centric Intelligent and Secure Systems June 2022 to date Department of Computer Science, Edge Hill University, UK

- ☑ Programme Leadership: Leading a team of five cyber security staff team
- ☑ Regularly lead outreach activities
- ☑ Founding Chair of HCIS²: Human-Centric Intelligent and Secure Systems
- ☑ Teaching undergraduate and postgraduate courses focused on Cyber Security and Forensics: Forensic Computing (final year, undergraduate); Introduction to Security (second year, undergraduate); Research and Development Methods (final year, undergraduate) and Research Methods (Masters)
- ☑ Designing and developing new modules and updating new modules: Infrastructure Security, System Security and Research Methods
- ☑ Module Leader: Forensic Computing (final year – undergraduate); Research Methods (Masters – leading a team of four colleagues) and Practical Data Analysis (Masters)
- ☑ Personal Academic Tutorship, Quality Assurance and Enhancement in Academic Processes
- ☑ External Examiner: University of Bedfordshire (Majan College, Oman)
- ☑ Independent Endpoint Assessor (Degree Apprenticeship – Level 7) – Northumbria University
- ☑ Administrative activities: focused on learning & teaching; research & development & management using tools such as Employee Expenses for financial management, E-Val for validation and accreditation processes, Student Information Desk (SID) Admin for student support, eFinancials for budget management, MyView for staff management, Budget Monitoring and Salary Costing Database for financial control, eRecruitment system, Tableau and other administrative platforms to streamline human resource, management and academic processes
- ☑ Advising Govt. of Pakistan on Several Issues
- ☑ Internationalization: established collaboration with an international partner
- ☑ Curriculum design and development, orientation and induction activities,
- ☑ PI of two funded projects funded by British Council
- ☑ Delivering induction and offer holder talks and taster sessions: BSc Computing (Networks, Cyber Security & Forensics), PG Induction – Academic Skills
- ☑ CyberFirst Bursary Scheme - NCSC - Student Orientation
- ☑ Deliver keynote and invited talks
- ☑ Member Teaching Quality Unit
- ☑ Supervising undergraduate and postgraduate students including PhD students
- ☑ Active National and International Research Collaboration
- ☑ Mentoring junior academic and postdoc staff
- ☑ Research activities such as chairing different stage PhD Vivas from different faculties within the university and member of research committee
- ☑ Elected as a member of Security and Privacy of CE Hardware and Software Systems (SPC) Technical Committee (TC) by IEEE CTSoc Consumer Technology Society
- ☑ Member: (i) Health Research Institute; (ii) Member Cardiorespiratory Research Centre; (iii) Centre for Data and Complex Systems Research; and (iv) Centre for Intelligent and Visual Computing.
- ☑ PhD Thesis Examiner & Viva Chair
- ☑ Grants Reviewer UKRI - EPSRC and British Council ISPF
- ☑ Head Invigilator (Computer Forensics and other exam)
- ☑ Serve on PhD GTA and Postdoc – Research Assistant shortlisting and interview panels
- ☑ ISC² Academic Peer-to-Peer Networking Group Member
- ☑ Other Scholarly and Academic Activities

Senior Lecturer in Cyber Security

Nov 2019 to June 2022

Cyber Security, Faculty of Computing Engineering and Science, University of South Wales, UK

- ☑ Teaching undergraduate and postgraduate courses (modules) focused on Cyber Security – currently teaching Practical Application Security (research focused - postgraduate); Security Management (Practice-focused); Network Security (hands-on focused - postgraduate); Security and System Administration (hands-on focused – undergraduate (final year)) and Security and Threat Management (hands-on focused – undergraduate (final year))
- ☑ Module Leader: Practical Application Security (MSc) and Security Management (MSc)

- ☒ Member of teaching and research team of National Cyber Security Centre (NCSC) certified trailblazing Academic Centers of Excellence (Won Gold Standard in 2020 – among only six universities of UK): Cyber@USW – the Department (University) also consecutively won (for three years) the award of the Cyber University of UK
- ☒ Member of QR Funding Strategic Implementation Group – A university wide committee chaired by Head of Research & Innovation Services
- ☒ Designed and developed high quality learning materials and detailed assessments for several modules such as Security and Threat Management; Security and System Administration and Network Security
- ☒ Writing and bidding for applied research grants
- ☒ Supervising undergraduate and postgraduate students including PhD students
- ☒ Member of COVID 19 Outbreak Expert Database of UK Parliament
- ☒ Active international research collaboration
- ☒ Delivering invited lectures
- ☒ Personal Academic Coaching
- ☒ Conducting research
- ☒ Member of conference organizing committees including Steering Committee Chair of international conferences
- ☒ Internal and External examiner of PhD theses
- ☒ Other academic and scholarly activities
- ☒ National (UKRI UK, EPSRC UK) and International Grant Reviewer (NSF, Pakistan and Higher Education Commission) Research Fellow (Cyber Security and Machine Learning) Oct 2018 to Nov 2019 Department of Computer Science, University of Surrey, United Kingdom
- ☒ Working on a large multimillion Euros EU funded project on Secure and Private Smart Grids with 17 partners from seven countries across the Europe
- ☒ Design and analysis of trust establishment and evaluation methods
- ☒ Design and analysis of anomaly detection methods
- ☒ Collaborating with international collaborators across Australia, Pakistan, Europe and USA
- ☒ Participating in international telcos, preparing documents and providing feedback on documents created by collaborators
- ☒ Providing feedback/ review on reports, hypotheses, experiment design and other phases of research process to other researchers/ reviewers involved in the project

Assistant Professor

Feb 2016 to Oct 2018

Department of Computer Sciences, Quaid-i-Azam University, Pakistan (No. 1 ranked Public Sector University – inside top 400-500 band of universities worldwide)

- ☒ Course Convenor and instructor of several PhD (CS) Courses such as Information Privacy and Access Control, Information Security, and Peer to Peer Systems
- ☒ Course Convenor and instructor of several M Phil (CS) Courses such as Information Privacy and Access Control, Information Security, Peer to Peer Networks, Distributed Systems
- ☒ Course Convenor and instructor of several MS (IST) Courses such as Web Services, Information Security, Information Privacy and Access Control
- ☒ Course Convenor and instructor of MSc (CS) Course, namely, Rapid Application Development (focused on web application development)
- ☒ Course Convenor and instructor of several BS (CS) Courses such as Introduction to Cyber Security, Operating System, Information System, Introduction to Web Services
- ☒ Curriculum and course development
- ☒ Conducting/ supervising research on Network and Data Security, Information Privacy and Security, web services, mobile agent based distributed systems, formal modelling and several other domains
- ☒ Supervised 9 postgraduate research students
- ☒ Supervised over 15 undergraduate project students
- ☒ Laboratories Coordinator – managing a team of seven persons including Technical Officers, Lab Assistants, and Lab Attendants
- ☒ Supervising design and rolling out new department website: cs.qau.edu.pk
- ☒ Redesigning network cabling infrastructure of the department according to the current environment
- ☒ Member PhD (CS), MPhil (CS), MS (IST), MSc (CS) admission committees
- ☒ Guest member in department's board of studies meetings
- ☒ BS (CS) programme accreditation activities
- ☒ Member of Selection Panel of Islamabad High court (public sector organization) for the recruitment of IT staff (2018)
- ☒ PhD and MPhil/ MS examiner of several universities such as Pir Mehr Ali Shah Arid Agriculture University, COMSATS Institute of Information Technology, Capital University of Science and Technology, and Shahid Zulfiqar Ali Bhutto Institute of Technology
- ☒ Elected Executive Council member of Academic Staff Association of QAU for the year 2017-2018
- ☒ Chair Organizing Committee formed by University to organize inter faculty-staff and employees cricket tournament of 22 teams
- ☒ PhD Comprehensive Exam Committee (S2017, F2017) - co-opted member

- ☒ BS (CS) Projects committee member (S2017, F2017)
- ☒ Coordinates study trip of students in collaboration with industry partners
- ☒ Coordinating minutes of BoS December 2016, minutes of semester planning committee, minutes of course allocation
- ☒ Designing five new courses

Lecturer

Aug 2015 to Jan 2016

School of Information and Communication Technology, Griffith University, Queensland, Australia

- ☒ Course Convenorship at undergraduate level
- ☒ Use of multimedia and ICT technologies to record and multicast lectures and other teaching materials
- ☒ Supervision of a large teaching team (6 members)
- ☒ Lecturing courses at Master and Bachelor levels
- ☒ Conducting collaborate sessions (online – virtual classroom)
- ☒ redesigning undergraduate courses by integrating Bloom's taxonomy in assessment items
- ☒ e-teaching to two large cohorts of students (around 250 and 175 students)
- ☒ Integration of online tools to facilitate teaching and learning processes
- ☒ Face-to-face teaching to a large cohort of students (around 300 students)
- ☒ Marking of Quizzes, assignments, and final examination papers
- ☒ Preparation of Labs and Lab manuals
- ☒ Creation and management of course website in accordance with the course profile
- ☒ Managing online discussion board to facilitate bidirectional communication between students and teaching team
- ☒ Examining presentations of students
- ☒ Providing formal and informal feedbacks to students on their course assessment items
- ☒ Performing research on Networks & Security, formal verification of communication protocols, mobile agent-based distributed systems, sensor networks, anomaly (intrusion) detection, cross-layer protocols

Instructor

Feb 2012- July 2015

School of Information and Communication Technology, Griffith University, Queensland, Australia

- ☒ Demonstrating labs of courses at Master and Bachelor levels
- ☒ Conducting workshops of courses at Master and Bachelor levels
- ☒ Course content preparation
- ☒ Course website development and management through Blackboard Learning Management System
- ☒ Teaching courses
 - Network Security
 - Computer Systems and Networks
 - Introduction to Information Systems
- ☒ Marking of Quizzes, assignments, and final examination papers
- ☒ Preparation of Labs and Lab manuals
- ☒ Creation and management of websites of courses in accordance with the course profiles
- ☒ Examining presentations of students
- ☒ Providing formal and informal feedbacks to students on their course assessment items
- ☒ Scored statistical mean score of 4.6 out of 5 in student experience of teaching report in Computer Systems and Networks – 1007ICT course
- ☒ Scored statistical mean score of 4.2 out of 5 in student experience of teaching report in Network Security – 3413ICT course

Lecturer

March 2011- To - Jan 2012

Faculty of Computer Science, King Khalid University, Abha, Aseer, Kingdom of Saudi Arabia

- ☒ Teaching multiple computer science courses
- ☒ Member of examination committee
- ☒ Preparing date-sheets, allocating rooms and allocating invigilators in rooms
- ☒ Coordinating in resolving students' issues which arise during examination process
- ☒ Coordinating and performing re-checking of examination scripts
- ☒ Coordinating in updating examination policies
- ☒ Member of research committee
- ☒ Coordinator of Computer Networks and Advanced Computer Networks
- ☒ Creating and updating course policies (course folder) of Computer Networks courses
- ☒ Developing course curriculums and resources
- ☒ Integration of Learning Management System (LMS) in teaching – online course management
- ☒ Student mentoring

- ☒ Coordination with concerned staff, Head of Department and Dean regarding degree accreditation process
- ☒ Application/ software support in laboratories
- ☒ Conducting labs and workshops
- ☒ Coordinating with course coordinators to revise and update course profiles

Lecturer and Project Manager (Research Centre)

Oct 2010 - To - Feb 2011 Faculty of

Information Technology, University of Central Punjab, Lahore, Punjab, Pakistan

- ☒ Teaching research focused Network Security course at postgraduate level
- ☒ Teaching computer science courses at postgraduate level
- ☒ Performing research in different areas of Computer Networks
- ☒ Supervising Masters level research in Network Security
- ☒ Coordinating BS (CS) final projects
- ☒ Providing administrative support to MS (CS) theses and PhD dissertations
- ☒ Course curriculum and resource development
- ☒ Course profile maintenance
- ☒ In charge Research Centre (Faculty of Information Technology)
- ☒ Maintained the Research Centre hardware by liaising with hardware vendors
- ☒ Coordination with industry partners to ensure industry standards of BS (CS) final projects
- ☒ Preparation of schedules and arrangement of BS (CS) projects, MS (CS) theses, and PhD dissertations defences
- ☒ Seminars coordination
- ☒ Integration of electronic course material using Blackboard Learning Management System

Lecturer and System Administrator

Oct 2008 – To - Sep 2010 PAF Fazaia

College, Lower Topa, Murree, Punjab, Pakistan

- ☒ Teaching and coordinating different courses such as, Fundamentals of Computers, C, GW Basics, Data Communications & Networks, Corel Draw & other Graphics Tools, E-commerce / Web Page Designing, Database Management Systems and Operating Systems
- ☒ Upgrading and smooth running of computer labs
- ☒ Intra-campus computer networking
- ☒ Deployment and smooth running of School Information Management System (SIMS) which includes following modules, HRM module, student progress module, accounts module, procurement and inventory module, library and laboratory module
- ☒ Generation of automated cellular reports for updating parents regarding educational status of their children
- ☒ Inter-campus e-testing of students over the corporate-level Wide Area Network
- ☒ Provision of intra-campus internet services
- ☒ Teachers training in use of computing resources and internet for teaching
- ☒ Administration staff training in information and communication technology resources
- ☒ Lab staff training in computer labs management
- ☒ Library staff training regarding centralized library database management system
- ☒ Provide assistance to Principal of College in information and communication technology related matters
- ☒ Coordination in provision of information and communication technology services in annual intra-campus principal conference
- ☒ Provision of information and communication technology services in annual and other events in the college

Network and Hardware Engineer

July 2003 – To - Oct 2008 Webcom Pvt.

Ltd. Islamabad, Pakistan

- ☒ Design and analysis of logical networks for routing protocols using the OPNET modeler before on-field deployment
- ☒ Configuration of e-mail and database servers
- ☒ Linux-based Installation and configuration of DNS, user accounts, HTTP, passwords, TCP, swap partitions, public key authentication, SSH port forwarding and remote servers' access
- ☒ Review of daily logs for unauthorized access and usage
- ☒ Managing and updating access policies
- ☒ Resolving technical issues in Microsoft operating systems and Red Hat Linux operating system
- ☒ Hands on experience on switching and routing devices
- ☒ Installation of Local Area Networks
- ☒ Perform layer 2 and 3 configurations and troubleshooting
- ☒ Network management
- ☒ Linux-based system administration
- ☒ Deployment of server systems and workstations
- ☒ Troubleshooting of server systems and workstations

- ☒ Office automation and document processing
- ☒ Database backup and restoration
- ☒ Periodic review and re-allocation of ICT resources based on need and supply principle
- ☒ Coordinate in provision of ICT consultancy services to the client organizations
- ☒ Windows server domain and Active Directory maintenance and upgrades using Microsoft windows 2000, 2003 and 2008
- ☒ Training of new staff
- ☒ Performance analysis of deployed networks

Consultant (Cyber Security - Research & Development - Higher Education)

Expertise areas include but not limited to the following domains:

- ☒ External Examiner
- ☒ Apprenticeship Independent Assessor
- ☒ Cyber Security
- ☒ SIEM Solutions and Security Operations Centre
- ☒ Higher Education - University Teaching (online/ blended/ face-to-face)
- ☒ Curriculum Development
- ☒ Information Assurance and Risk Assessment
- ☒ Research & Development
- ☒ IoT and Industrial IoT
- ☒ Career Professional Development Training
- ☒ Cyber Security Training
- ☒ IT and IOT consultancy
- ☒ Security Architect
- ☒ Information Management System
- ☒ Information and Communication Technology
- ☒ Cyber laws, IT and Cyber Governance, Security Management
- ☒ IT and Cyber Standards (ISO 27000, 27001, 27002, 27003, 22305, 27006, 27007, 27008, 27031, 27033, 27035, 27036, 27037, 28000, 18043, 10012, others)
- ☒ Cyber Essentials and Cyber Essentials Plus

Contributions to Administration and Management

Leadership and Administrative Activities: In addition to the activities mentioned throughout this CV and supporting document, in summary I have a proven track record of providing **leadership to IT staff and teaching teams**. I have led both academic and staff teams in different occasions. I am currently leading a BSc Programme (Networks, Cyber Security and Forensics). In several funded projects, I have directly line managed or co-line managed RAs and Postdocs. In one of my past positions, I have **managed an IT team of six members** to oversee the IT related matters in the Department. My **mentorship to junior teaching staff** illustrates my commitment to fostering the professional growth and development of my colleagues. This mentoring role reflects my willingness to share knowledge, offer guidance, and help junior faculty members excel in their teaching roles.

I have demonstrated ability to provide academic leadership through various facets of his academic activities. I have served as both Principal Investigator (PI) and Co-Investigator (Co-I) in several funded research projects. These roles require strong leadership skills to oversee the research direction, coordinate activities among team members, and ensure project objectives are met effectively. Leading or participating in successful funding applications requires leadership skills to articulate research visions, build consortia, and secure financial support for research projects. My involvement in various funded projects demonstrates my leadership in securing research funding. I am currently leading a research group focused on Human-Centred Intelligent and Secure Systems. This indicates my capability to provide leadership in guiding the research direction, mentoring junior researchers, and fostering collaboration within the group.

My involvement in research projects with international collaborations showcases my ability to establish and maintain relationships with researchers and institutions worldwide. This demonstrates **leadership in fostering global research networks and promoting collaboration across borders**. My publication record includes a range of papers published in reputable academic journals and conferences, indicating leadership in conducting rigorous research, disseminating findings, and contributing to academic discourse in my field. My involvement in editorial roles, such as guest editorials for journals, suggests recognition and leadership in the academic community. Serving as an editor involves providing leadership in shaping the content and direction of academic publications.

As a Principal Investigator and Co-Investigator on numerous projects, I **contributed to curriculum development** by integrating cutting-edge research findings into educational programs. This involvement includes updating course content, introducing new modules or courses, and aligning curriculum with industry trends and technological advancements. I have organised and lead academic events such as colloquiums, workshops, and conferences aimed at fostering knowledge exchange, professional development, and networking opportunities among researchers, academics, and students. These events contributed to the intellectual growth of the academic community and facilitated interdisciplinary collaboration. Through my involvement in various interdisciplinary research projects and collaborations, I ensured interdisciplinary collaboration within academic settings. I have **advocated for diversity and inclusion within academic environments**, promoting equitable access to educational opportunities

and fostering a culture of inclusivity and belonging.

I possess **digital leadership and administrative skills** that enable me to navigate through a wide array of technological tools and platforms with ease and efficiency. My competency spans various software applications, database management systems, and analytical tools commonly used in academic research, project management, and administrative tasks. I use productivity software such as Employee Expenses for financial management, E-Val for validation and accreditation processes, Student Information Desk (SID) Admin for student support, eFinancials for budget management, MyView for staff management, Tableau for student tracking, Budget Monitoring and Salary Costing Database for financial control, eRecruitment system, Tableau and other administrative platforms to organise data, collaborate with colleagues, and streamline workflows.

Teaching Profile

Over the years, I have taught several undergraduate and postgraduate courses in the broader domain of the information and communication technology. I have also taught several research-focused postgraduate courses. I have supervised and supervising over 100 undergraduate project and postgraduate research students that have produced a number of publications.

Teaching Philosophy

Each student is an integral part of the classroom learning environment and each individual’s learning curve is different from others. Maximizing the learning experience of each student by maintaining an appropriate pace, covering sufficient course depth, and achieving course learning outcomes are therefore challenging tasks. I believe early observations, communication, and learning and teaching feedbacks play vital role to create a good learning and teaching environment. Then continuous and careful monitoring of the performance of students and appropriate suggestions to overcome their weaknesses can help them to learn most out of the course content. Also, the interactive and dynamic (that is, lecture plus activities-based) class environment makes students confident to express their knowledge and to better synthesize the core concepts of a course.

Student Feedback

An unsolicited feedback from my recent student at Edge Hill University, UK.
“Dear Dr. Usman,

I hope this email finds you well. I visited Edge Hill University today to personally thank you for your exceptional teaching and guidance throughout the Research Methods and Practical Data Analysis modules, but unfortunately, I missed you.

Your expertise and enthusiasm significantly impacted my understanding of research methodologies and data analysis techniques. I'm grateful for:

1. Your clear explanations and engaging lectures

2. Practical examples and hands-on exercises

3. Feedback and guidance on assignments and projects
- In Research Methods, your teaching helped me:

1. Develop a solid foundation in research design and methodology

2. Understand the importance of ethical considerations

3. Critically evaluate research studies

In Practical Data Analysis, your guidance enabled me to:

1. Master data analysis techniques using various tools

2. Interpret complex data insights effectively

3. Apply data-driven decision-making skills

Thank you for believing in us and pushing us to excel. Your influence has enhanced my research and data analysis skills, confidence, and academic achievements.”

Courses/ Modules Taught

Sr. No.	Course Title
1	Information Privacy and Access Control (module leader, Quaid-i-Azam University, 2016-2018)
2	Information Security (module leader, Quaid-i-Azam University, 2016-2018)

3	Network Security (module leader, University of South Wales, 2019-2022)
4	Cyber Security (module leader, Quaid-i-Azam University, 2016-2018)
5	Security and System Administration (module leader, University of South Wales, 2019-2022)
6	Web Services (module leader, Quaid-i-Azam University, 2016-2018)
7	Information System (module leader, Griffith University, Australia, 2015-2016)
8	Distributed Systems (module leader, Quaid-i-Azam University, 2016-2018)
9	Rapid Application Development (module leader, Quaid-i-Azam University, 2016-2018)
10	Computer Systems and Networks (module tutor, Griffith University, Australia, 2013-2015)
11	Computer Networks (module leader, King Khalid University, 2011 - 2012)
12	Advanced Computer Networks (module leader, King Khalid University, 2011 - 2012)
13	Data communications (module leader, King Khalid University, 2011 - 2012)
14	Communications and Networking (module leader, King Khalid University, 2011 - 2012)
15	Fundamentals of Computers (module leader, King Khalid University, 2011 - 2012)
16	Introduction to Computer Programming (C, C++) (module leader, Fazaia College, 2008 - 2010)
17	Introduction to Information Systems (module leader, Fazaia College, 2008 - 2010)
18	Database Design (module leader, Fazaia College, 2008 - 2010)

19	Database Management Systems (module leader, Fazaia College, 2008 - 2010)
20	Operating System (module leader, Quaid-i-Azam University, 2016-2018)
21	E-Commerce (Website Designing) (module leader, Fazaia College, 2008 - 2010)
22	Software Engineering (module leader, Fazaia College, 2008 - 2010)
23	Security and Threat Management (module leader, University of South Wales, 2019-2022)
24	Forensic Computing (module leader, Edge Hill University, 2022- 2024)
25	Research & Development Methods (module tutor, Edge Hill University, 2022- 2024)
26	Practical Data Analysis (module leader, Edge Hill University, 2023 - 2024)

Research Focused Modules of Interest

Sr. No.	Course Title
1	System Modelling and Analysis Using Petri Net Theory
2	System Modelling and Analysis Using Behavior Engineering
3	System Modelling and Analysis Using Fuzzy Logic

Other Major Courses of Interest

Sr. No.	Course Title
1	Wireless Networks
2	Integrated Services Over Packet Networks
3	Advanced Operating Systems
4	Linux-based System Administration
5	Microsoft Windows-based System Administration

- ☒ Research supervision at postgraduate level
- ☒ Undergraduate projects supervision
- ☒ Teachers training at college level
- ☒ Staff training

Teachers Training and Related Workshops

- 1) Enhancement and Quality for Educators (ENQUIRE) 2024, The Quality Assurance Agency for Higher Education (QAA) England: April 2024.
- 2) ASIThe Role of the Academic Tutor at EHU, 2023.
- 3) E-Val, Edge Hill University, UK, October 2022.
- 4) Demystifying Governance (Academic Governance), Edge Hill University, UK, October 2022.
- 5) Researcher Development session: Demystifying Open Access, Edge Hill University, UK, October 2022.
- 6) KEF/ REF External Relations and Research Development Introduction, Edge Hill University, UK, November 2022.
- 7) Transgender Awareness and Inclusivity Training, Edge Hill University, UK, December 2022.
- 8) Personal Tutor Training, Edge Hill University, UK, December 2022.

- 9) Reflection and Observation of Practice Scheme, University of South Wales, UK, February 2022.
- 10) TSL@USW, HEA UK Fellowship claim course, University of South Wales, UK, December 2020/January 2021.
- 11) USW DEAL: Principles into Practice, 2021.
- 12) Lab management (LanSchool application) workshop – 25 February 2015 - Griffith University, Australia
- 13) Blackboard 9.1 management workshop - 17 February 2015 - Griffith University, Australia
- 14) Course Profile Workshop - 27 November 2014 - Griffith University - conducted by MS. Georgina, a Curriculum Consultant at Griffith University, Australia
- 15) Australian (South Eastern Queensland) Computing Education Conventicle - 30 October 2013 - Griffith University, Australia
- 16) One-week teachers training workshop - June 2009, PAF Fazaia College, Pakistan
- 17) One-week teachers training workshop - July 2010, PAF Fazaia College, Pakistan

Research Profile

Dr Muhammad Usman is currently investigating several contemporary research issues in the domains of security and privacy, the internet of things, artificial intelligence, web services, data analytics, formal and statistical modelling and analyses and human-centred intelligent systems. The research in these domains has a potential role to play in many critical national infrastructure applications. On the technical side, this work integrates many areas such as formal modelling and analysis of communication and security protocols, design and analysis of energy-efficient systems, wireless sensor networks, and cross-layer optimization. Prior to this work, he published numerous papers in different research domains such as authentication in mobile ad hoc networks, key management schemes, seamless and secure vertical handoff, group mobility in ad hoc networks, and energy efficiency protocol design in resource-constrained networks.

Publications:

Submitted:

1. Jeremiah, D., Rafiq, H., Ta V-T., and **Usman, M.**, Awais, M., & Raza, M. NIOM-DGA: Nature-Inspired Optimised ML-based Model for DGA Detection”, *Computers & Security, Elsevier* (revision submitted - under review).
2. Onwuajuese, O. S., Rafiq, H., **Usman, M.**, & Ta, V-T., Evasion Aware Android Malware Detection System based on Optimised KNN Model. *IEEE Transactions on Information Forensics and Security*, (under review).
3. Hameed, **A.**, **Usman, M.**, Zia, T., Haq, Q. E., & Irshad, A. FeD-AD: A Federated Learning based Anomaly Detection and Recovery Prediction using Temporal Transformers for Drone Forensics. *Internet of Things*, (under review).
4. Hameed, A., **Usman, M.**, Abbas, H., Zia, T., & Kashif, A. B. Ranking of Internet-of-Medical-Things Services using Service Association Factor. *IEEE Transactions on Industrial Cyber-Physical Systems* (under review).
5. Rafiq, H., Aslam, N., Issac, B., Jeremiah, D., Onwuajuese, O. S., **Usman, M.**, and Ta, V-T. Adversarial Attacks and Defenses on ML-based Android Malware Detectors. *ACM Computing Surveys* (to be submitted in April 2025).
6. **Usman, M.**, Alsaadi, M., Begum, A. U., Irshad, A., Soni, M., Mekhmonov, S. and Haider, S., Adaptive Block-Based Holographic Video Streaming for IoT Consumer Electronics, *IEEE Transactions on Consumer Electronics* (2025, under review).
7. **Usman, M.** and Irshad, A. SAC-DeV: Secure Access Control for Drone-Assisted Edge Computing in e-VANETs, *IEEE Transactions on Intelligent Transportation Systems*, (2025, under review).

Published:

Book / Editorials

8. Irshad, A., **Usman, M.**, Reddy, A. G., Chaudhry, S. A., and Mahmood, K. Agentic AI-driven Cyber Physical/IoT ecosystem leading to secure and dependable intelligent transportation system, Special Issue Proposal, *IEEE Internet of Things Journal* (2025, under review)
9. Biswas, K., Chowdhury, M. J. M., & **Usman, M.** (Eds.). (2024). Guest Editorial: Blockchain of

Things: Benefits, Challenges and Future. *Sensors*, (ISSN 1424-8220).

10. Irshad, A., Shafiq, M., Chaudhry, S. A., & **Usman, M.** (2022). Guest Editorial: Security Hardened and Privacy Preserved Vehicle-to-Everything (V2X) Communication. *Security and Communication Networks*, volume 2022.
11. Jolfaei, A., **Usman, M.** et al. (2021). Guest Editorial: Computational Intelligence for Human-in-the-Loop Cyber Physical Systems. *IEEE Transactions on Emerging Topics in Computational Intelligence*.
12. **Usman, M.**, et al. (2020). Guest Editorial: Security, Reliability, Privacy, and Quality in Industrial Automation and Control. *IEEE Transactions on Industry Applications, IEEE*.
13. **Usman, M.**, Muthukkumarasamy, V., Wu, X.-W., & Khanum, S. (2018). Mobile Agent-Based Anomaly Detection and Verification System for Smart Home Sensor Networks. Springer.

Journal

14. Khan, S., Khan, M., Khan, M. A., Luo, F., **Usman, M.**, Irshad, A., Wu, K., Wan, L., Khan, M. A., PBatch: Pseudonym Certificate Batch Authentication for Cooperative Intelligent Transportation Systems, *IEEE Transactions on Intelligent Transportation Systems*, 2025, (accepted).
15. Imran, M., Amjad, M., Afzal, MK, **Usman, M.**, Kim, BS. (2025). Investigating Low Latency Communications with Link-Layer Rate of NOMA under Nakagami-m Fading Channel in Finite Blocklength Regime, *IEEE Access*.
16. Irshad, A., Aljaedi, A., Bassfar, Z., Jamal, S. S., **Usman, M.**, and Chaudhry, S. A. (2024). FA-SMW: Fog- Driven Anonymous Lightweight Access Control for Smart Medical Wearables," in *IEEE Internet of Things Journal*,
17. Ahmad, T., **Usman, M.**, Murtaza, M., Benitez, I. B., Anwar, A., Vassiliou, V., Irshad, A., Li, X. J., & Al- Ammar, E. A. (2024). A Novel Self-Calibrated UWB based indoor localization systems for context-aware applications. *IEEE Transactions on Consumer Electronics*.
18. Aljaedi, A., Siddique, S., Satti, M. I., Alharbi, A. R., Alotaibi, M., & **Usman, M.** (2023). Underpinning Quality Assurance: Identifying Core Testing Strategies for Multiple Layers of Internet-of-Things-Based Application. *Sustainability*.
19. Praveen, N., Juneja, K., Rashid, M., Almagrabi, A. O., Sekaran, K., Ramalingam, R., & **Usman, M.** (2023). Hybrid gated recurrent unit and convolutional neural network-based deep learning mechanism for efficient shilling attack detection in social networks. *Computers and Electrical Engineering*, 108.
20. Hasan, K., Chowdhury, M. J. M., Biswas, K., Ahmed, K., Islam, M. S., & **Usman, M.** (2022). A blockchain- based secure data-sharing framework for Software Defined Wireless Body Area Networks. *Computer Networks*, 211.
21. Rauf, B., Abbas, H., **Usman, M.**, Zia, T., Iqbal, W., Abbas, Y., & Afzal, H. (2022). Application Threats to Exploit Northbound Interface Vulnerabilities in Software Defined Networks. *ACM Computing Survey*, 54(6).
22. Carie, A., **Usman, M.**, Marapelli, B., Bedru, H. D., & Biswas, K. (2021). A Human-in-the-Loop Probabilistic CNN-Fuzzy Logic Framework for Accident Prediction in Vehicular Networks. *IEEE Sensors Journal*, 21(14), 15496–15503.
23. Chowdhury, M. J. M., **Usman, M.**, Ferdous, M. S., Chowdhury, N., Harun, A. I., Jannat, U. S., & Biswas, K. (2021). A Cross-Layer Trust-based Consensus Protocol for Peer-to-Peer Energy Trading Using Fuzzy Logic. *IEEE Internet of Things Journal*, 9(16).
24. Javed, A., **Usman, M.**, Rehman, S. ur, Khan, M., & Haghighi, M. S. (2021). Anomaly Detection in Automated Vehicles Using Multistage Attention-based Convolutional Neural Network. *IEEE Transactions on Intelligent Transportation Systems*, 22(7).
25. Mahmud, K., **Usman, M.**, Sindhu, M. A., Jolfaei, A., & Srivastava, G. (2021). Closing the Loop in Feedback Driven Learning Environments Using Trust Decision Making and Utility Theory. *IEEE Transactions on Emerging Topics in Computational Intelligence*, 5(1).
26. Irshad, A., **Usman, M.**, Chaudhry, S. A., Bashir, A. K., Jolfaei, A., & Srivastav, G. (2020). Fuzzy-in-the- Loop Driven Low Cost and Secure Biometric User Access to Server. *IEEE Transactions on Reliability*, 70(3).
27. Waheed, N., He, X., Ikram, M., **Usman, M.**, Hashmi, S. S., & Usman, M. (2020). Security and Privacy in IoT Using Machine Learning and Blockchain: Threats & Countermeasures. *ACM Computing Surveys*.

28. Hameed, A., **Usman, M.**, & Maqbool, O. (2020). Privacy Preserved Ranking of Industrial Sensing Services Using Topological Information. *IEEE Transactions on Industry Applications*, 56(4), July.
29. Irshad, A., **Usman, M.**, Chaudhry, S. A., Naqvi, H., & Shafiq, M. (2020). A provably secure and efficient authenticated key agreement scheme for Energy Internet based Vehicle-to-Grid technology framework. *IEEE Transactions on Industry Applications*, 56(4), January.
30. Irshad, A., Ashraf, S., Shafiq, M., **Usman, M.**, & Kumari, S. (2019). An improved biometric Multi-Server Authentication Scheme for Chang et al. protocol. *Information Technology and Control*, 48(2).
31. Irshad, A., Chaudhry, S. A., Shafiq, **M.**, **Usman, M.**, Asif, M., & Ghani, A. (2019). A provable and secure mobile user authentication Scheme for Mobile Cloud Computing Services. *International Journal of Communication Systems*, 32(14), September.
32. Irshad, A., Naqvi, H., Chaudhary, S. **A.**, **Usman, M.**, Shafiq, M., Mir, O., & Kanwal, A. (2018). Cryptanalysis and improvement of a Multi-Server Authenticated Key Agreement by Chen and Lee's Scheme. *Information Technology and Control*, 47(3).
33. Mahmud, K., & **Usman, M.** (2018). Trust Establishment and Estimation in Cloud Services: A Systematic Literature Review. *Journal of Network and Systems Management*, 27(2), pp. 489-540, October.
34. Shafiq, M., Ahmad, M., Irshad, A., Gohar, M., **Usman, M.**, Afzal, M. K. I., Yu, H., & Choi, J.-G. (2018). Multiple Access Control for Cognitive Radio-Based IEEE 802.11ah Networks. *Sensors*, 18, June.
35. Irshad, A., Naqvi, S. H. A., Chaudhary, S. A., Raheem, S. T., Kumari, S., Kanwal, A., & **Usman, M.** (2018). An efficient and secure design of multi-server authenticated key agreement protocol. *The Journal of Supercomputing*, 74(9), pp. 4771-4797, July.
36. Batool, Z., **Usman, M.**, Saleem, K., Al-Wadud, M. A.-A., Amin, F.-e., & Al-Eliwi, A. (2018). Disease- Disease Association Using Network Modeling: Challenges and Opportunities. *Journal of Medical Imaging and Health Informatics*, 8, pp. 1-10.
37. Irshad, A., Chaudhary, S. A., Kumari, S., **Usman, M.**, Mahmood, K., & Faisal, M. S. (2017). An improved lightweight multi-server authentication scheme. **International Journal of Communication Systems*, 4 July.
38. **Usman, M.**, Muthukkumarasamy, & Wu, X.-W. (2015). Mobile Agent-based Cross Layer Anomaly Detection in Smart Home Sensor Networks. *IEEE Transactions on Consumer Electronics*.
39. **Usman, M.**, Muthukkumarasamy, V., & Wu, X.-W. (2015). Specifications and Validation of Agent-based Anomaly Detection and Verification System for Resource Constrained Networks. *Journal of Networks*.
40. **Usman, M.**, Muthukkumarasamy, V., & Wu, X.-W. (2014). A Resource-Efficient System for Mobile Agent- based Anomaly Detection and Verification in Wireless Sensor Networks. *Journal of Networks*, 9(12), pp. 3427-3444.
41. Khanum, S., **Usman, M.**, & Alwabel, A. (2012). Mobile Agent Based Hierarchical Intrusion Detection System in Wireless Sensor Networks. *International Journal of Computer Science Issues (IJCSI)*, 9(1).
42. Irshad, E., Noshairwan, W., Shafiq, M., Khurram, S., Irshad, A., & **Usman, M.** (2011). Performance Evaluation of Group Mobility in Mobile Ad hoc Networks. *International Journal of Future Generation Communication and Networking*, 4(1), pp. 13-22.
43. Irshad, A., Noshairwan, W., Shafiq, M., Khurram, S., Irshad, E., & **Usman, M.** (2008). Security Enhancement in MANET Authentication by checking the CRL Status of Servers. *International Journal of Advanced Science & Technology (IJAST)*, Korea.

Conference/ Book Chapter

44. Udeze, S., Rafiq, H., Jeremiah, D., **Usman, M.**, and Ta, V-T. (2024). Image-Based Android Malware Detection Using Deep Learning. 16th International Conference on Global Security, Safety & Sustainability, ICGS3-24, Newcastle, UK.
45. Onyeoma, C. F., Rafiq, H., Jeremiah, D., Ta V-T., and **Usman, M.**, (2024) "Credit Card Fraud Detection Using Deep Neural Network with Shapley Additive Explanations" International Conference on Frontiers of Information Technology (FIT), Islamabad, Pakistan.
46. Jaffar, I., **Usman, M.**, & Jolfaei, A. (2019). Security Hardening for Implantable Cardioverter Defibrillators. Special Session on Security Hardening of Industrial IoT Systems in 20th IEEE international conference on industrial technology, IEEE-ICIT, Melbourne, VIC, Australia, pp. 1173-1178.
47. Zia, A., & **Usman, M.** (2018). Elective Learning Objects Group Recommendation Using Non-Cooperative Game Theory. IEEE International Conference on Frontiers of Information Technology (FIT), Islamabad, Pakistan, pp. 194-199.
48. Lone, S., & **Usman, M.** (2018). E-HOSA: Extended-Hybrid Optical Switch Architecture for Cloud Computing Data Centers. IEEE International Conference on Frontiers of Information Technology (FIT), Islamabad, Pakistan, pp. 53-58.

49. Hameed, A., & **Usman, M.** (2017). Sum-of-the-Parts Valuation-based Scheduling of Parallel Applications over Global Grid. International Conference on Frontiers of Information Technology (FIT), Islamabad, Pakistan, pp. 133-138.
50. Mehmood, K., & **Usman, M.** (2017). Dynamic Trust Evaluation of Learning Objects. International Conference on Communications Technologies (ComTech - 2017), Rawalpindi, Pakistan, pp. 195-200. **(Best Paper Award)**.
51. **Usman, M.**, Muthukkumarasamy, V., Wu, X.-W., & Khanum, S. (2014). Anomaly Detection in Wireless Sensor Network: Challenges and Future Trends. In Security for Multihop Wireless Networks, Auerbach publications, Taylor & Francis Group, USA, pp. 273-304.
52. **Usman, M.** (2014). Agent-enabled Anomaly Detection in Resource Constrained Wireless Sensor Networks. IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM), Sydney, Australia.
53. **Usman, M.**, Muthukkumarasamy, V., & Wu, X.-W. (2013). Formal Verification of Mobile Agent Based Anomaly Detection in Wireless Sensor Networks. IEEE 38th conference on Local Computer Networks (LCN Workshops), Sydney, Australia.
54. Biswas, K., Muthukkumarasamy, V., Sithirasanen, E., & **Usman, M.** (2013). Efficient Clique Based Clustering and Routing Mechanism in Wireless Sensor Networks. 9th IEEE International Wireless Communications and Mobile Computing Conference (IWCMC), Sardinia, Italy.
55. **Usman, M.**, Muthukkumarasamy, V., Wu, X.-W., & Khanum, S. (2012). Securing Mobile Agent Based Wireless Sensor Network Applications on Middleware. IEEE 12th International Symposium on Communication and Information Technology (ISCIT), Gold Coast, Australia.
56. **Usman, M.**, Muthukkumarasamy, V., Wu, X.-W., & Khanum, S. (2012). Wireless Smart Home Sensor Networks: Mobile Agent Based Anomaly Detection. IEEE 9th conference on Ubiquitous intelligence and Computing (UIC), Japan.
57. Haleem, A., & **Usman, M.** (2011). An Optimal Mutual Authentication Scheme in GSM Networks. IEEE International Conference on Information and Communication Technologies (ICICT), Karachi, Pakistan.
58. Irshad, A., Gilani, S. M., Khurram, S., Shafiq, M., Khan, A. W., **Usman, M.**, et al. (2010). Hash-chain based peer-peer key management and establishment of Security Associations in MANETS. IEEE International Conference on Information and Emerging Technologies (ICIET), Karachi, Pakistan.
59. **Usman, M.**, Khanum, S., Noshairwan, W., Irshad, E., & Irshad, A. (2010). Performance Evaluation of Authentication Certificate Based Seamless Vertical Handoff in GPRS- WLAN. In High Performance Computing and Applications (HPCA), LNCS, Springer-Verlag, pp. 384-389.
60. Khanum, S., **Usman, M.**, Hussain, K., Zafar, R., & Sher, M. (2010). Energy-Efficient Intrusion Detection System for Wireless Sensor Network Based on MUSK Architecture. In High Performance Computing and Applications (HPCA), LNCS, Springer-Verlag, pp. 212-217.
61. Irshad, A., Noshairwan, W., Shafiq, M., Khan, A. W., **Usman, M.**, & Irshad, E. (2009). Certificate Chain based Authentication in MANETS using 4th generation Technologies. IEEE 2nd International Conference on Computer Science and Information Technology (ICCSIT), Beijing, China.
62. Irshad, A., Noshairwan, W., Rashid, M., Gilani, S. M., Irshad, E., & **Usman, M.** (2009). Authentication of Nodes among Different Symmetric Key Groups in MANETs using 4G Technologies. IEEE International Conference on Computer Engineering and Technology (ICCET), Singapore.
63. Irshad, A., Shafiq, M., Rahman, A., Khurram, S., **Usman, M.**, & Irshad, E. (2009). A Secure interaction among nodes from different MANET Groups using 4G technologies. In Proceedings of the IEEE International Conference on Emerging Technologies (ICET), pp. 476-481, Rawalpindi, Pakistan.
64. **Usman, M.**, Noshairwan, W., Gilani, M., Irshad, A., & Irshad, E. (2008). Seamless vertical handoff using Authentication Certificate in GPRS-WLAN tightly coupled integrated networks. In Proceedings of the 4th IEEE International Conference on Emerging Technologies, pp. 199-204, Pakistan.
65. Irshad, E., Noshairwan, W., **Usman, M.**, Irshad, A., & Gilani, M. (2008). Group Mobility in Mobile Ad hoc Networks. In WWW/Internet, IADIS, pp. 1-5, Germany.

PhD Supervision:

Completed:

1. PhD: Shoaib Ahmed, Sentiment Analysis of COVID-19 Vaccination Tweets, Islamia University Bahawalpur, Pakistan (Co-supervision with Professor Dost Muhammad Khan), (20 – 23) - Completed.
2. PhD: Javed Iqbal, Software Quality Assurance of Web-based Medium Scale Projects Using Agile Umbrella, Islamia University Bahawalpur, Pakistan (Co-supervision with Professor Dost Muhammad Khan), (20 – 23) - Completed.
3. PhD: Dr Nazar Waheed, Security and Privacy Risks in Web-based Chatbots, University of Technology Sydney, Australia (19-21) – co-supervised for over one year – completed in 2023.

Ongoing:

4. PhD: Amna Noor, Hierarchical Chatbot with Topic Modelling and Generative Hierarchical Neural Networks, Edge Hill University, UK, (23- 26).
5. PhD: Syed Muzammil Ali, AI-Based Human Activity and Vital Sign Monitoring Using Hybrid RF Sensing, Edge Hill University, UK, (23- 26).
6. PhD: Jawad Babar Pirzada, building a Crowd Counting Model Using Deep Learning, Islamia University Bahawalpur, Pakistan (Co-supervision with Professor Dost Muhammad Khan), (20 – 25).
7. PhD: Zahid Aslam, Fake News Detection Using Blending Machine Learning Classifiers, Islamia University Bahawalpur, Pakistan (Co-supervision with Professor Dost Muhammad Khan), (20 – 25).

Other Postgraduate Students (Completed):

1. MSc: Liam Molyneux, Cyber Issues of Remote Working During COVID-19, a Comprehensive Analysis and Risk Assessment Framework, Edge Hill University, UK, (22 – 23).
2. MSc: Nathan Johnson, Countering Deepfake Risks: A Cybersecurity Framework for Mitigating the Risks of Deepfakes, Edge Hill University, UK, (22 – 23).
3. MSc: Jake Eacott, The Cyber Security Implications of the Remote Working Transition During COVID-19, Edge Hill University, UK, (22 – 23).
4. MSc: Sukhwinder Kaur, The Role of Cloud Computing in Education: Benefits, Challenges, and Future Prospects, Edge Hill University, UK, (22 – 23).
5. MSc: ESOSA Igunbor, Secure Cloud Services: An Architecture for Policy Customisation and Trustworthiness, Edge Hill University, UK, (22 – 23).
6. MSc: Abdul Kareem Siddiqui, Analyse logs of Intrusion Detection System (IDS) / Intrusion Prevention System (IPS) & Honeypot for network security support, Edge Hill University, UK, (22 – 23).
7. MSc: LAKSHMI ANUSHA GANAPATHI, Analysing the impact of edge computing on cloud-based IoT, (with Dr Hassan), Edge Hill University, UK, (22 – 23).
8. MSc: Kim Smith, Design and Analyse a Socio-technical Business Risk and Vulnerability Assessment Process for Industrial Control Systems, first supervisor with Dr Ian Wilson, University of South Wales, UK, (19 – 22).
9. MSc: Ihsan Al Hamoud, Design and Analysis of Internet-of-Things Driven Intelligent Aquaponics System, Second supervisor with Dr Eurfyl Davies and Dr Christopher Tubb, University of South Wales, UK, (19 – 22).
10. MSc: An Investigation into digital evidence artefacts resulting from the use of cloud services with an added focus on Amazon Web Services, Google Cloud Platform and Microsoft Azure on Windows 10 Operating System - Luke Jones (with Gareth Davies), University of South Wales, UK, (19 - 20).
11. MSc: Security measures on cloud computing - Yaqoob Alrayes (with Peter Eden), University of South Wales, UK, (19 - 20).
8. MPhil: Key Exchange Protocols in Internet of Things Applications – Zartaj Ali, Quaid-e-Azam University, Pakistan, (S'18 – F'19).
9. MPhil: Design and Analysis of Optical Interconnection System for Cloud Computing Data Centers - Samina Lone, Quaid-e-Azam University, Pakistan, (F'17 – S'19).
10. MPhil: Design and Analysis of a Data Leakage System – Seema Raishad, Quaid-e-Azam University, Pakistan, (F'17 – S'19).
11. MPhil: Key Exchange Protocols in Cloud Computing – Maryam Mahmood, Quaid-e-Azam University, Pakistan, (S'17 – S'19).

12. MPhil: Ranking of Sensing-as-a-Service in Internet of Things - Aroosa Hameed, Quaid-e-Azam University, Pakistan, (F'17 – S'18).
13. MPhil: Probabilistic Judgments in Analytic Hierarchy Process for Web Services Ranking – Noor Alam, Quaid-e- Azam University, Pakistan, (F'16 – S'18).
14. MPhil: Game Theory-based Learning Object Selection – Asad Zia, Quaid-e-Azam University, Pakistan, (F'16 – S'18).
15. MPhil: Risk Factors-based Network Analysis of Non-Communicable Diseases – Zahra Batool, Quaid-e-Azam University, Pakistan, (F'16 – S'17).
16. MPhil: Utility Theory-based Trust Evaluation of Learning Objects – Kanwal Mahmud, Quaid-e-Azam University, Pakistan, (F'16 – S'17).

Development Projects by Undergraduate Students

MSc (CS)/ BS (CS) (Selected Only)

- ☒ Forensic Computing – Eve Baden, 22 – 23
- ☒ An Evaluation of Cryptocurrency and Blockchain in regard to operational, transactional and safety purposes - George Betham, 22 – 23
- ☒ Security Protocol Design for Autonomously Connected Vehicle/ Vehicle Cloud - Megan Bradbury, 22 – 23
- ☒ Blockchain-based Patient Data Management System - Cameron Challoner, 22 – 23
- ☒ Securing Mobility - Zeb Clark-Wolstenholme, 22 – 23
- ☒ Pen testing or designing a defence/ attack method via practical means - Conner Croft, 22 – 23
- ☒ Blockchain-based loyalty card system – Joseph Druggan, 22 – 23
- ☒ Security Protocol Design and (Verification) for Autonomous Connected Vehicles - Connor Edwardson, 22 – 23
- ☒ What are the possible repercussions and solutions of an unsecure business network - Connah Wilson, 22 – 23
- ☒ How the security of systems keeps them from threats – Matthew Henry, 22 – 23
- ☒ Security in Blockchain - Alistair Hoyle, 22 – 23
- ☒ The analysis of security and vulnerabilities of storing data in cloud – Elizabeth Scully, 22 - 23
- ☒ Evaluating the Security and ethical risks of IOT devices & Smart Home technology - James Morgan, 20 – 21
- ☒ Cloud Security - Threat prevention and security deployment - David Jarslett, 20 – 21
- ☒ Seeing Under the Mask - Facial Recognition - Matthew Hopes, 20 – 21
- ☒ Is it possible to track illegal activity on the dark web through the means of cryptocurrency tracing - Evan Jones, 20 – 21
- ☒ Security Consultancy - Marius Stene Johansen, 20 – 21
- ☒ How safe is your password - understanding theory around secure passwords and user attitudes towards them? - Mikkel Andreas Aronsen, 20 – 21
- ☒ Botnets of compromised IoT devices in the smart home environment - Patryk Marek Przybocki, 20 – 21
- ☒ Privacy and Security of Electronic Voting using Blockchain technology - Elliot Douglas Rihan, 20 – 21
- ☒ How to Develop a Blockchain Using the Ethereum IDE Platform - Connor McGoggy, 20 – 21
- ☒ The best defenses against social engineering - Thomas Sandnes, 20 – 21
- ☒ Mobile Browser Forensic Investigation - Jesper Næss
- ☒ IoT Attack Vectors – The vulnerable world of IoT device - Gjermund Fougner Haugen, 20 – 21
- ☒ Researching and implementing methods for AV evasion in a customised Remote Access Trojan
- ☒ Researching and implementing methods for AV evasion in a customised Remote Access Trojan - Carl Isak Lundbye Haugan (with Ahmed Elmesiry), 20 – 21

- ☑ Has WPA3 made consumer routers more secure? - Glenn Bygdevoll Allred (with Mamoun Qasem), 20 – 21
- ☑ Drone Security/ Drone Hacking - Ben Jeffery, 20 – 21
- ☑ To develop a Forensic readiness Playbook for when your data is held at Ransom? - Matthew Jones (with Gareth Davies), 20 – 21
- ☑ Importance of using Open-Source Intelligence effectively in Law Enforcement - Declan Christy (with Peter Eden), 20 – 21
- ☑ Investigation of how digital evidence plays a part in criminal investigations - Savannah Hill-Andrews (with Kim Smith), 20 – 21
- ☑ Developing a training course for the appropriate tools that can be used for Open-Source Intelligence - Sindre Homman (with Gareth Davies), 20 – 21
- ☑ Conducting reconnaissance and vulnerability scanning from covert device, remote controlled by phone/tablet - Ernst Hetland Larsen (with Ahmed Elmesiry), 20 – 21
- ☑ Cyber Security - Jack Williams (with Adnrew Ballamy), 20 – 21
- ☑ Hacking of invasive heart pacemakers - Stian Rønning (with Peter Eden), 20 – 21
- ☑ The interconnected world challenges and opportunities - Vegard Kristensen (with Mamoun Qasem), 20 – 21
- ☑ Telehealth - Mubeen Sarwar (F'17 – S'18)
- ☑ Calendar Analyzer– Bisma Tariq (S'17 – F'17)
- ☑ Taste Skewer – Nadia Liaqat (S'17 – F'17)
- ☑ Employee Tracker – Waheed Ahmed (S'17)
- ☑ Friends Band - Mahvish Ghazi (S'17)
- ☑ Student Learning Experience - Iqra Shabeer (S'17)
- ☑ Visualization of Real-World Applications using Distributed Synchronization and Scheduling Algorithms - Saba Bashir (F'16)
- ☑ Family Tree - Zeeshan Shahid (F'16)
- ☑ Transport Management System: Quaid-I-Azam University - Muhammad Umar (S'16)
- ☑ Smart Contracts – Raja Ammad Iftikhar (F'17, S'18)
- ☑ Development of Intra-Lab Data Leakage Prevention System – Danish Alam (F'17, S'18)
- ☑ Cricket Club Hub - Abdul Rahman (F'16, S'17)
- ☑ Academic Staff Association Hub - I - Hassan Farooqi (F'16, S'17)
- ☑ Academic Staff Association Hub - II - Azmatullah (F'16, S'17)
- ☑ Quaid-I-Azam University Community Guild - I - Luqman Anjum (F'16, S'17)
- ☑ Quaid-I-Azam University Community Guild - II - Shahroz Jamil (F'16, S'17)

PhD Examination (non-exhaustive list)

- ☑ The Origin Study: A biomechanical comparison of functional recovery after custom or conventional total knee replacement, Edge Hill University, (PhD Final Viva Chair), UK 2024.
- ☑ Classification of Maize Fungal Diseases using Deep Learning Technique, The Islamia University of Bahawalpur, (External Examiner), Pakistan 2024.
- ☑ Individual differences in social cognition: Does loneliness affect our ability to identify and utilise non-verbal cues? Edge Hill University, 2023 (PhD Progression Viva Chair), UK, 2023.
- ☑ Multiple Patient Simulation for Preparing Student Nurses for Undergraduate Clinical Practice: An Educational Design Research Study, Edge Hill University, 2023 (PhD Registration Viva Chair), UK, 2023.
- ☑ Short-lived newspapers: Reassessing success and failure in the nineteenth-century press, Edge Hill University, 2023 (PhD Progression Viva Chair), UK, 2023.
- ☑ A multi-tier trust-based security management system for vehicle ad-hoc network communication, Edge Hill University, (Internal Examiner) UK, 2023.
- ☑ Mapping Multiple Interdependencies, University of South Wales, UK, 2022.
- ☑ Networks, Federation University, Australia, 2021.
- ☑ Designing a Trust Management System for Malicious Node Detection and Prevention in Internet of Things, National University of Science and Technology, Pakistan, 2021
- ☑ Resource Efficient Multi-Dimensional Cache Management Strategy in Content-Centric Network, CUST, Pakistan, 2018.
- ☑ Artificial Intelligence-based Congestion Detection Mechanism in Opportunistic Networks, SZABIST, Pakistan, 2018.

Recent Invited Talks and Advisory Roles

- ☑ Invited talk: Unmasking the Unseen: Innovations in Document Forensics, The Sixth International Conference on the

Arab Society for Forensic Sciences & Forensic Medicine Conference, Saudi Arabia, December 2023.

- ☒ Invited talk: Securing Healthcare Infrastructure, Lancashire Cyber Partnership, organized by University of Central Lancashire, United Kingdom, November 2023.
- ☒ Keynote talk: Federated Learning-based Anomaly Detection and Recovery Prediction for Stratosphere Drone Forensics in Fifth International Conference on Advances in Electrical and Computer Technologies, India, 2023.
- ☒ Invited talk: Interdisciplinary Research for Societal Impact, Islamia University of Bahawalpur, Pakistan, 2023.
- ☒ Invited talk: Industry and Societal Needs Driven Research for Sustainable Development, University of Wah, Pakistan, 2023.
- ☒ Keynote talk: Role of AI in Addressing Societal Challenges Focused on Healthcare, Smart Farming, Green Energy and Societal Issues, AI-Driven Interdisciplinary Research Hackathon, PAK-UK Academic Mobility Programme, Jointly Funded by British Council and Higher Education Commission of Pakistan, COMSATS University Wah Campus, Islamabad, 2023.
- ☒ Keynote talk: Cyber Security in Healthcare Services in IEEE International Conference on Electrical, Electronics, Information and Communication Technologies (ICEEICT 2023), India, 2023.
- ☒ When Computing and Cyber Security Meet Healthcare Services, Healthcare Research Institute, Edge Hill University, UK, March 2023.
- ☒ Inaugural speech and keynote talk: Consensus Mechanisms in Internet of Robotics Things for Extreme Environments" at fifth IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT 2023) - co-sponsored by IEEE USA and IEEE MADRAS Sections, India, February 2023.
- ☒ Keynote talk: "Artificial Intelligence Driven Drones Forensics for Sustainable and Cognitive Infrastructure", ICAECT, India, January 2023.
- ☒ Advisory Member to Ministry of Planning, Government of Pakistan, 2023.
- ☒ Advisory Board Member of Centre of Excellence in Cyber Security, VIT Bhopal University, India.
- ☒ External examination panel for PhD and MPhil for Islamia University of Bahawalpur, Pakistan
- ☒ Trust-based Consensus Protocols, UK Cyber Security PhD Winter School, University of Surrey, UK, 2022.
- ☒ Blockchain-enabled Peer-to-Peer Energy Trading, jointly hosted by EU IoT Council, Europe and The Islamia University of Bahawalpur, Pakistan on March 25, 2021.
- ☒ Threats and Countermeasures in Connected Toys, British Science Week Events by University of South Wales, March 8, 2021.
- ☒ Critical National Infrastructure, Cyber CNI, Mines-Télécom Institute, France, February 2, 2021.

Editorial Services, Invited Reviewer & Member

Role	Journal/ Conference
Editor	• Guest Editor: <i>IEEE Transactions on Consumer Electronics</i>, Enhancing Edge-based Consumer Electronics Devices and IoT, 4th Quarter 2025. • Guest Editor: <i>Sensors</i> Special Issue on Blockchain of Things: Benefits, Challenges and Future Directions, 2022. • Guest Editor: <i>IEEE Transactions on Emerging Topics in Computational Intelligence</i> Special Issue on "Computational Intelligence for Human-in-the-Loop Cyber-Physical Systems" Scheduled to appear in late 2021. • Guest Editor: Security Hardened and Privacy Preserved Vehicle-to-Everything (V2X) Communication, Security and Communication Networks, November 2021. • Lead Associate Guest Editor for a Special Issue of <i>IEEE Transactions on Industry Applications</i> on Security, Privacy, Trust, and Access Control in Industrial Applications, published in June/ July 2020.
Member Editorial Board	• Editorial board member of <i>Smart Science Journal</i> by Taylor & Francis

General/ Co-General Chair	- Higher Education Leadership Summit, Edge Hill University, UK, November 2023. - Quality Assurance and Standard Setting: Validation, Moderation & Best Academic Practices; Edge Hill University, UK, November 2023. - Digital Learning: Tools, Techniques & Practices; Edge Hill University, UK, November 2023. - Intelligent Visual Computing Research; Edge Hill University, UK, November 2023. - Data and Complex Systems Research; Edge Hill University, UK, November 2023. - Interdisciplinary Research; Edge Hill University, UK, November 2023. - Knowledge Transfer Partnership and Apprenticeships, Edge Hill University, UK, November 2023. - AI-Driven Interdisciplinary Research Hackathon, PAK-UK Academic Mobility Programme, Jointly Funded by British Council and Higher Education Commission of Pakistan, COMSATS University Wah Campus, Islamabad, 2023. - International AI Colloquium on Interdisciplinary Research on rural health, smart farming, renewable and green energy, socio-economic uplift of minority, PAK-UK Academic Mobility Programme, Jointly Funded by British Council and Higher Education Commission of Pakistan, NUST Islamabad, 2023. - International AI Symposium on rural health, smart farming, renewable and green energy, socio-
	economic uplift of minority, PAK-UK Academic Mobility Programme, Jointly Funded by British Council and Higher Education Commission of Pakistan, Islamia University Bahawalpur, 2023.
Steering Committee	- Workshop on Artificial Intelligence Driven Advancements in Cyber Security and Digital Forensics in 2023 The 5th European Symposium on Computer and Communications, Manchester UK, April 13 - 15, 2023 (Organizing Chair) - IEEE ICOIDT2 2021 (Steering Committee Chair) 7th International Conference on Cyber Security, Privacy in Communication Networks (ICCS) 2021 – IAASSE - (Organizing Chair) 8th International Conference on Cyber Security, Privacy in Communication Networks (ICCS) 2022 – IAASSE - (Organizing Chair)
Session Chair	- Springer - ICCS 2022: 8th International Conference on Cyber Security and Privacy in Communication Networks (ICCS) 2022 - Session Chair (Advances in AI/ ML/ Deep Learning) in International Conference on Computing and communication networks, ICCCN, Manchester UK, 2022 - ICCS 2021 (7th International Conference on Cyber Security, Privacy in Communication Networks)
Member TPC/ Organising Committee	- Organizer: Special Session on Beyond Boundaries: Interdisciplinary Perspectives on Vision, Language, and Ethical Considerations in AI-driven Clinical Diagnosis and Early Detection in 28th UK Conference on Medical Image Understanding and Analysis, Manchester, UK 2024. - Invited TPC member for the 8th Symposium on Distributed Ledger Technology, Springer, Australia, 2024 - The 2023 IEEE 98th Vehicular Technology Conference will be held Fall 2023 in Hong Kong - TPC member (Track: Spectrum Management, Green Communications, Services and Security) - ACM ASIACCS 2023 (Safety and security in intelligent transportation systems workshop) - ICCCN 2022 (The 31st International Conference on Computer Communication and Networks) - DependSys 2019 - First IEEE International Workshop on Internet of Services and Applications (IoS 2018) - Member Technical Program Committee - Member Technical Program Committee for ICCAIS-18 - IBCAST Reviewer - 2018 - Intellect 2017 - IBMSGs 2015 - UWCSN 2014 - ICCAIE 2010 - UbiCoNet 2010
Grant Reviewer	- Systemic AI Safety Grants Programme, AI Safety Institute, UK Government, EPSRC UKRI - Future Leaders Fellowships: Round 8 & 9, UKRI - UKRI, UK - British Council - Pakistan Science Foundation - Higher Education Commission, Pakistan

Reviewer	• IEEE Transactions on parallel and distributed systems • IEEE Transactions on Cybernetics • Journal of Systems and Software, Elsevier Journal • Journal of Networks and Computer Applications, Elsevier Journal • International Journal of Image and Data Fusion • IEEE Access • International Journal of Image and Data Fusion • ISBEIA 2014 • ISIEA 2013, ISCI 2013, BEIAC 2013 • BEIAC 2012, IAPEC 2012, APACE 2012, PECON 2012, SCORed 2012, ISBEIA 2012, ISCAIE 2012, ISCI 2012, ISIEA 2012, ISWTA 2012, • ICOS 2011, RFM 2011, SCORed 2011, ISCI 2011, ISIEA 2011, ISWTA 2011 • FEMnet 2010, ICCAIE 2010, Wi-UAV 2010, UbiCoNet 2010
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Membership of Organizations

Organization	Description
Computer Science Teacher Association (CSTA) endorsed by ACM Publishers	Membership no.: 1730769
Juniper Networks	• Member as an Internetworking Specialist (Enterprise Routing Track), • Internetworking Specialist (JUNOS Security Track), • Internetworking Associate (Enterprise Switching Track), • Internetworking Associate (Enterprise Routing Track)
Institute for Integrated and Intelligent Systems (IIIS), Griffith University	Member

Consultation Experience

- Member of Recruitment Panel of ICT executive and lower-level staff at Islamabad HighCourt, Pakistan
- Advisor to Recruitment Panels of several ICT and Cyber Security Companies
- Consultant Cyber Security, IT Support, R&D, Curriculum Design & Development, Higher Education

Skill	Description
Skills	
System and network Administration	- Linux-based System and Network Administration - Microsoft Windows-based System and Network Administration
Major networking protocols/ algorithms/ technologies/ servers	LAN, MAN, WAN, Intrusion Detection System, Telnet, http, https, ftp, VPN, BGP, Proxy Server, Linux Servers, Microsoft Windows Servers, Linux Netfilter, iptables, Virtualization (Type 1 and Type 2 hypervisors), Oracle Virtualbox, VMware, Cloud computing, Cross-layer optimization, Cross-layer Security, Formal requirement analysis through behaviour engineering and Petri net theory, Security risk assessment, Network and security report writing, Design, and analysis of networks
Computing tools	Latex, NS ₂ , OMNET++, Opnet modeler, Riverbed Modeler, Adobe, Corel Draw, Microsoft Visio, Blackboard, Learning Management System, Adobe Flash, MS Office, Inkscape, and GnuPlot
Programming languages	C, C++, VB, GW Basic, HTML, ASP, JSP, SQL, MATLAB
Consultancy	- Computer Networks and Security - System design and deployment - Performance analysis and tuning of computing and networking systems
Typing speed	50+ words per minute

Languages

Language	Proficiency
English	Professional working proficiency
Urdu	Native or bilingual proficiency
Punjabi	Native or bilingual proficiency
Arabic	Limited working proficiency
Potohari	Professional working proficiency
Saraiki	Limited working proficiency

Extra-Curricular Activities & Interests

I am a cricket enthusiast, and I have played club cricket. I was part of my school and college cricket and football teams, and I have won several certificates and trophies in these games. I am the president of a football club in Pakistan and have been an Executive Member (management) of a Cricket Club in the UK. I have been a captain of a Mid-week cricket team at Manchester Amateur Cricket League, where I led the T20 team to finish as runners-up in the league round and 2nd in the Cup games. I was a top-run scorer in the league a few years ago. In my leisure time, I like reading newspapers, books, online materials, and playing cricket.

Referees

1. Professor Vallipuram Muthukkumarasamy School of Information and Communication Technology, Griffith University, Gold Coast Campus, Australia email: v.muthu@griffith.edu.au

2. Professor Tanveer Zia

Head of Computer Science, University of Notre Dame, Australia email: tanveer.zia@ne.edu.au

3. Dr Xin-Wen Wu

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